ECNS 560 Term Project

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Throughout the rest of the semester, you will pursue an independent project to analyze data in an area of your own interest. This is your chance to show off all the skills you've learned in this course – I expect you to challenge yourself! It will be different from a typical research project in economics – we won't focus quite as much on causal inference, but more on descriptive analysis, as well as on the process you follow to get there. The final outputs will consist of a GitHub code repository, a written report, and a short peer review. You will meet with me as a team for early feedback and individually for a final project walk-through.

Teams. I strongly encourage you to work in teams of 2 students. The project may also be done alone, but expectations and grading standards will be the same. Teams of more than 2 students are not generally permitted. If you really want to work in a team of 3, you will need to meet with me to ask for special permission, and your team will be held to higher expectations. If you would like me to assign you to a partner, I am happy to play matchmaker – just email me.

Use of AI. You will use AI in your future work, but the purpose of this project is to get firsthand practice coding and fully using your own mind to work through problems. I strongly discourage AI in this project, but enforcement is up to you. Ultimately, you are fully responsible for all code you submit. You must understand every line and be able to explain it and defend every choice.

Stage 1: Topic and Data

Due Thursday, March 12 by class time, on GitHub.

Agree on a topic for your project with your partner. It can be anything relevant to economics, public policy, or business, but you should both find it interesting!

- You cannot choose the exact same topic as another team. If you hear that another team is working on a similar topic, talk with them and negotiate which team will work on which aspect of the topic (or just choose another topic).
- You *may* use the same topic and datasets that you are using for the project in ECNS 562. However, you may not submit the same visualizations or regression tables for the projects in both courses. You will need to find a different angle to the analysis for the two projects; the work needs to be distinct in ways that are more substantial than just a few small tweaks.

Think of a few potential research questions. Write down a few (2-4) well-defined research questions that you might be interested in answering in this project. A research question is a complete sentence that can be answered with the right data and assumptions. It is narrower and more precisely defined than a topic. Here are a few examples of good research questions:

1. (Descriptive) Has wealth inequality increased in the U.S. faster than in Europe since 2000?
2. (Causal) Does receiving Medicaid coverage reduce the risk of bankruptcy?
3. (Predictive) Can nighttime satellite imagery be used as a real-time indicator of economic activity?

Find data from at least two different sources that are related to your topic. Download them and make sure you're going to able to use them for analysis. The datasets should relate to each other in some way that will allow you to join (merge) them together, but you don't have to do that yet. All data should be publicly available and should come from the original (primary) source. They should not be pre-cleaned – a major objective of this project is for you to go through the process of downloading (or scraping!), importing, wrangling, and cleaning the data.

Create a public repository on GitHub for your project. *(You can always make your repo private after the semester if you prefer, but past projects on your GitHub account will look great to future employers.)* Add a document that describes your topic, motivate why it matters, and lists your research questions. Then, describe your datasets. Include the source URL, what kind of variables are contained in each dataset (no need for a full list), the timespan and spatial coverage of the datasets, and how the datasets are related to each other.

Submit the URL of your GitHub repo to Canvas, so I know where to find it. I will provide feedback, but you're also welcome to discuss your project with me in office hours.

Feedback 1: Project Planning Meeting

To be scheduled the week of March 24.

Each team will meet with me for 10-20 minutes to discuss your plans for the project. This meeting is not evaluative – it's for your benefit. I'll ask you to show me your Stage 1 submission and I'll do my best to give you some useful feedback on your initial plans.

Stage 2: Draft Exploratory Analysis

Due Thursday, April 9 by class time, on GitHub.

Wrangle and clean: Import your datasets into R, wrangle them into tidy format, and join them into a single data frame. Clean your data, keeping in mind our Data Cleaning Checklist. Your code should be heavily commented and follow the best practices that we discuss in class.

Explore: Conduct an exploratory analysis of your data. A good starting point is to focus on one outcome variable (or a small number of outcome variables) and investigate its unconditional distribution, then how it varies across geography, over time (or by month/week/day/hour), and/or according to other key categorical or quantitative variables in your data. (You may then want to show the distribution of and variation in those other key variables.) Apply transformations where appropriate, pay attention to missing data, and handle extreme values in a defensible way.

Visualize and communicate: Produce a set of compelling figures that show the most important things going on in your data. Provide context for the more formal analysis you will conduct in Stage 3. Try to give a provisional, visual answer to your research question. Tell a story with graphs!

- Each figure should have a clear, focused purpose. Think carefully about the principles of good data visualization we discuss in class. Ask yourself: Is this figure the very best way of conveying the information I am trying to communicate?
- As a very rough guideline, you might aim for 4-6 well-constructed figures, but the appropriate number will depend on your specific circumstances.
- Your visualizations should look professional, contain titles, labels, and captions, and follow the principles of good data visualization we discuss in class. There are limitless numbers of

cool things you can do with `ggplot2` beyond what we cover in class. See if you can impress me!
- Be precise with language: Avoid implying causal relationships between variables at this stage, since you're merely describing your data. You don't want your audience to make high-stakes decisions based on unwarranted interpretations!

Write up a report in R Markdown, containing:

- Your topic, motivation, and the description of your data that you already submitted (revised as necessary to reflect any changes).
- A couple of paragraphs explaining how you processed your data and why you made the choices you did. Identify the unit of analysis in your cleaned, merged dataset.
- A couple of paragraphs explaining any transformations you applied, how you dealt with any missing data or extreme values, and why.
- A few pages that present your visualizations and interpret them.

Focus on quality over quantity. Endless pages of graphs or text are not necessarily impressive. Make your report succinct but thorough. Your report should look neat and professional. Unlike the homework assignments, it should **not** contain R code or raw output. Instead, export your figures as image files and then embed them in your write-up. Use headings to organize the document, write clearly and straightforwardly but not informally, and check your spelling and grammar.

Collaborate on GitHub: If working with a partner, use GitHub as your platform for sharing files throughout the assignment. If working alone, still use it to store and back up your project as you work. **Each team member must make at least one substantive commit** to the repo. (The objective of this expectation is for each team member has some practice with GitHub. I will not examine the record of commits for other purposes, since the number of commits will not necessarily reflect the amount of effort and contributions of each team member.)

Organize your project repo, following the best practices we discussed in class, such as:

- Put raw data, code, cleaned data, and output in separate folders. Give all files and folders informative names.
- Thoroughly document your code with comments. Make it as easy as possible for a future collaborator or replicator to understand how to run all of your code and create the outputs.
- Do not use absolute filepaths in your scripts. Allow each user (including yourself) to define the location of the project folder on their own computer, by setting the working directory in the R console.
- Ordering your scripts with numbered filenames, or creating a "master" script that runs everything else with one click.

Commit and push a complete exploratory analysis to your GitHub repository. Make sure your repo includes:

- The R scripts you used to process your data and produce the visualizations. (These should be in standalone R scripts, not R Markdown documents. Do not include the code for everything you tried or looked at, only what's necessary to produce the results.)
- The final cleaned dataset as an object in a `.Rdata` file.
- The knitted report in **HTML or PDF format,** as well as the raw file(s) used to create it. (If you choose to submit it in HTML, make sure your repo includes all image or CSS files it depends on.)

Feedback 2: Peer Review

In class on Thursday, April 9.

The day Stage 2 is due, your team will swap projects with another team and complete the following steps:

First, attempt to reproduce the other team's project. Clone their GitHub repository to your computer, delete all output and temp files, and try to generate all output and results in their report.

Second, read the other team's draft exploratory analysis. Think about how effectively the analysis shows the reader what's going on in the data. Does the report clearly communicate what the team wants to do, and what they have done? Does the analysis seem appropriate for the research question? Does the analysis make appropriate decisions around handling data? Are the visualizations clear and effective?

Third, collaborate to write 1-2 pages of feedback for the other team. Your feedback can be written informally; I recommend bullet points. It should include:

1. **A short reproducibility report.** What worked? What didn't? What could have been easier?
2. **A score for reproducibility.** Rate the repository out of 10 points, where:
    - "10" means you couldn't imagine an easier project to reproduce.
    - "8" means there were minor hassles or errors but it worked.
    - "6" means there were major hassles or errors, but you were ultimately able to reproduce most of the contents of the report.
    - "4" means you were able to run substantial parts of the code but could not reproduce most of the report.
    - "2" means the repo had some data and code but you did not successfully run much of it during the time allotted.
    - "0" means you were not able to access any of the project inputs.
3. **Strengths of the draft.** What do they do well? What is already highly effective? (Make sure to balance your criticism with praise!)
4. **Constructive criticism.** What could be more effective? In what ways could the analysis be improved, and how? Try to offer constructive ideas alongside each piece of criticism.

Finally, your teams will meet to exchange reviews (and submit them to me). You will then have three more weeks to revise your project and improve it before it is graded.

Stage 3: Final Exploratory Analysis + Econometric Analysis

Due Thursday, April 30 by class time, on GitHub.

Once you have completed your exploratory analysis, you now understand your data well enough that you're ready for an econometric analysis. However, I do not expect this analysis to be a complete research project. Do not focus on finding a good identification strategy, or trying to convincingly answer a causal question (you will not be able to do so in the time allowed!). Instead, finalize a research question (or a small number of closely related questions) that your data can speak to. Then, conduct one of the following types of analysis:

1. **Descriptive regression analysis, as an end in itself.** If your research question(s) of interest is a descriptive question, use regression to help you quantify (and obtain standard errors for) some of the relationships that you found in the exploratory analysis.
   a. Write down 1-3 regression models that you could use to shed light on one or more of your research questions using your available data you have. No need to get fancy – a simple OLS or fixed effects regression is fine. I won't prohibit you from trying something else you've learned in econometrics, but start with something simple. Explain all notation in your regressions and why they are appropriate to your situation.
   b. Run your regressions and display the estimate(s) of interest in a table. (A figure is optional.)
   c. Interpret your results. Put the coefficient(s) into complete sentences of English to explain what the regression told you. Explain what they mean. What new facts do we learn about the world from the relationships (or lack of relationships) you find? Since your research question is only descriptive, be *very* careful not to claim or even imply any causal interpretation of your results.
2. **Descriptive regression analysis, as a first step toward causal inference.** If your research question is a causal question, do steps a-c above, plus a couple more things:
   a. Interpret your results through the lens of causal inference. How plausible would it be to interpret your results causally? Describe the limitations to doing so. What are some likely sources of bias, in what direction are they likely to push, and how concerned should we be about them? It's often helpful to describe what the ideal experiment would be, and then compare your situation to that experiment.
   b. Say a few words about how you might proceed from here, if you were going to more convincingly answer your causal research question.
3. **Predictive analysis.** If your research question is a predictive question, try using machine learning tools to conduct a predictive analysis.
   a. Choose one of the variables in your cleaned dataset to be the target of prediction. You should be able to explain why a policymaker, business executive, or another member of the general public might find it useful to be able to predict this outcome variable.
   b. Use other variables in your cleaned dataset to predict the outcome variable as best you can. Follow all the correct ML practices: Set aside a test set; train and predict using one or more ML algorithms; tune your model(s) using appropriate cross-validation, and finally test your performance with the held-out data. (Don't cheat!!! Really, don't touch the test set until you are done tweaking your model. If you change your model after seeing how it does in the test set, then you are reporting dishonest performance numbers.)
   c. Explain your methods: What learning method did you use? What parameters did you tune? What method did you use for tuning?
   d. Interpret your results: How did you measure performance/success? How did your model(s) perform? What do you think limited the performance? What did you learn in the process? It may be effective to create visualizations to accompany your interpretation.
4. **Prediction in service of causal inference.** If you want to go all out, you can try answering a causal research question using double/debiased machine learning (which I won't cover until November 21, but I can provide you materials in advance if you ask). The idea is that

you can use predictive models to reduce omitted variables bias in a regression that you want to interpret causally. If you choose to pursue this, you will need to follow all applicable steps from each of the other types of analysis.

Then:

Add a section to your report describing your econometric analysis and the results. Use headers like Motivation, Methods, Results. Include all information required above for the type of analysis you chose. Update the first section of your report so that it describes the research question you landed upon and why it's important or interesting. Be sure to explain why you chose the type of analysis you did: What makes your research question either a descriptive, causal, or predictive question, and not either of the other two?

Commit and push your project to GitHub. Continue to follow best practices for organizing your project. Ensure your final repo includes all scripts, inputs, and outputs involved in the econometric analysis, as well as the updated report.

Feedback 3: Project Review Meeting

To be scheduled the week of April 30.

Each student (not each team) will meet individually with me for 15-20 minutes to walk through your project. I'll clone your repository and ask you to walk me through your project, with a focus on your own individual contributions. We'll first go through the scripts and then the report. I just want you to tell me about what you did, and I'll likely have questions for you along the way. This is your chance to show off your work – I'm excited to see it!

Stage 4: Team Evaluation

Due Thursday, May 7 by 2 pm, on Canvas.

Unless you worked individually, submit a short evaluation of whether you and your team member(s) distributed work equally or if the work was unequal. If it was unequal, describe the contributions of each team member and whether you believe you should receive a different grade than the other team member(s).

Grading

Project grades will be based on the following categories:

- Peer review of another project (10 points).
- Reproducibility score from peer review (10 points).
- Code and data (30 points).
    - Style, coding practices, file organization.
    - Data importing and wrangling.
    - Data cleaning.
- Exploratory analysis (30 points).
    - Data exploration.
    - Visualizations.
    - Interpretation and effectiveness.

- Econometric analysis (20 points).
    - Technical execution.
    - Interpretation and effectiveness.
- Challenging yourself (10+ points).

An “A” grade means your project exceeds expectations. It:

1. Demonstrates a wide range of skills covered in this course.
2. Carefully follows best practices from nearly all topics in this course.
3. Challenges yourself in a meaningful way (i.e., you went beyond the assignment minimum expectations).

A “B” grade means your project meets expectations. It fulfills the requirements and shows a range of skills from this course. It may not precisely follow every best practice but it gets the job done.

A “C” grade means your project demonstrates some skills from this course but has significant problems or omissions that limit its effectiveness.

Key areas of best practices include:

1. Project/file organization
2. Abstraction and modularity
3. Commenting and documentation
4. Joins and primary keys
5. Data cleaning checklist
6. Missing data
7. Extreme values
8. Variable transformations
9. Visualization principles
10. Descriptive vs. causal vs. predictive interpretation
11. Separating training and testing data (if relevant)

Examples of “challenging yourself” include:

1. Using data that requires extraordinary effort to assemble, merge, and/or clean.
2. Obtaining data using unusual methods, such as webscraping, APIs, or OCR.
3. Applying spatial tools as an integral part of the analysis.
4. Conducting a predictive analysis using ML tools.
5. Setting up an especially thoughtful research design for causal inference.
6. Working with datasets so large they require special computing resources or tools.
7. Taking on a conceptually sophisticated question that relies on nontrivial economic theory to interpret the results.